

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

Attack identification: The detected attack is a SYN Flood, which is a specific type of Denial of Service (DoS) attack.

Traffic analysis: The network log shows an abnormally high volume of TCP SYN requests originating from a single unknown IP address.

Technical evidence: The web server is receiving requests to initiate a connection (SYN), but the attacker never completes the process, leaving the server in an infinite waiting state until its resources are exhausted.

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Section 2: Explain how the attack is causing the website to malfunction

Description of the attack and symptoms

This attack abuses the TCP connection process (the "three-way handshake").

The attacker sends the SYN packet, the server responds with a SYN-ACK and reserves memory for the connection, but the attacker never sends the final ACK.

As a result, the server accumulates thousands of "half-open connections," consuming all of its processing capacity and memory.

Impact on the organization's network

Unavailability: Employees cannot access the sales page to search for vacation packages, which halts business operations.

Timeout error: Legitimate customers receive a "connection timeout" message because the server is so busy handling fake requests that it ignores the real ones.

Consequences for the company

Loss of revenue: By not being able to close sales or promotions, the agency directly loses money.

Reputational damage: Lack of access to the website generates distrust in customers.

Recommendations and next steps

Immediate mitigation: The source IP has been blocked in the firewall and the server was restarted to release the hung connections.

Long-term solutions:

Implement SYN cookies to prevent the server from reserving resources before completing the handshake.

Configure rate limiting on the firewall to block IPs that send too many requests in a short time.

Install an intrusion prevention system (IPS) that automatically detects flooding patterns.